形式化语言与自动机理论 - 作业 4

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1. 给出接受以下语言的 PDA.

   $L \subseteq \{a, b, c\}^*$ and $|a| = |b| + |c|$

答案：

定义 PDA $M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$，其中

$Q = \{q_0, q_1\}$

$\Sigma = \{a, b, c\}$

$\Gamma = \{Z_0, a, b, c\}$

$F = \{q_1\}$

$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$

$\delta(q_0, b, Z_0) = \{(q_0, bZ_0)\}$

$\delta(q_0, c, Z_0) = \{(q_0, cZ_0)\}$

$\delta(q_0, a, a) = \{(q_0, aa)\}$

$\delta(q_0, a, b) = \{(q_0, \varepsilon)\}$

$\delta(q_0, a, c) = \{(q_0, \varepsilon)\}$

$\delta(q_0, b, a) = \{(q_0, \varepsilon)\}$

$\delta(q_0, b, b) = \{(q_0, bb)\}$

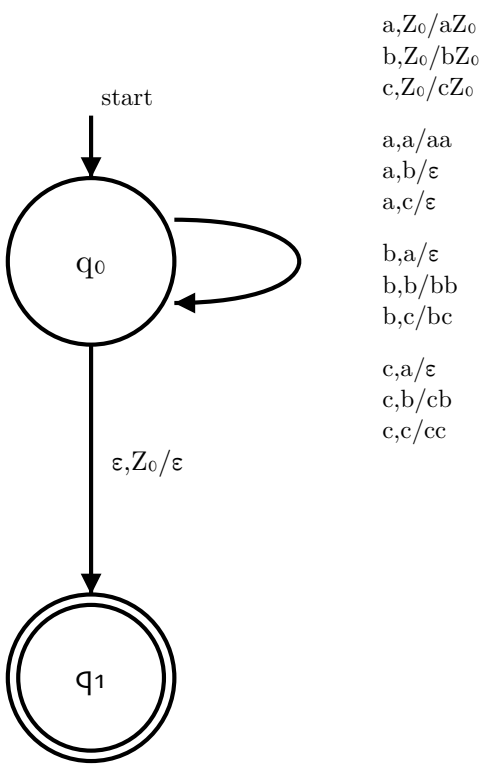
$\delta(q_0, b, c) = \{(q_0, bc)\}$

$\delta(q_0, c, a) = \{(q_0, \varepsilon)\}$

$\delta(q_0, c, b) = \{(q_0, cb)\}$

$\delta(q_0, c, c) = \{(q_0, cc)\}$

$\delta(q_0, \varepsilon, Z_0) = \{(q_1, \varepsilon)\}$


start
q0
q1
ε,Z0/ε
a,Z0/aZ0
b,Z0/bZ0
c,Z0/cZ0
a,a/aa
a,b/ε
a,c/ε
b,a/ε
b,b/bb
b,c/bc
c,a/ε
c,b/cb
c,c/cc